

PHY240 PS2, problem 7: verifier card

Standing waves (counter-propagating waves, E-B misalignment, energy sloshing, zero net energy flow)

How to use this card. It does not contain the answers. For each part it lists tests you can run on *your own* result: plug in the numbers given, take the limits, try the special cases. A result that passes every test is probably right; a result that fails one has gone wrong somewhere, and the "wrong turns" list names the usual places. Check the conventions first: the tests assume exactly the symbols and signs defined below.

The two waves are $\mathbf{E}_1(x, t) = E_{10} \cos(kx - \omega t) \hat{y}$ and $\mathbf{E}_2(x, t) = E_{20} \cos(kx + \omega t + \phi) \hat{y}$.

Conventions

- Phases: $a = kx - \omega t$ for wave 1 (moving toward $+\hat{x}$), $b = kx + \omega t + \phi$ for wave 2 (moving toward $-\hat{x}$). Both electric fields are along $\hat{y}$.
- $\mathbf{B}$ follows from Faraday's law, $\partial_t B_z = -\partial_x E_y$, with the integration constant set to zero; for a single plane wave this gives $\mathbf{B} = \hat{k} \times \mathbf{E}/c$ with $\hat{k}$ the direction of travel of *that* wave.
- $\mu_0 = 1/(\epsilon_0 c^2)$, so $u_B = B^2/2\mu_0 = \epsilon_0 (cB)^2/2$ and $\mathcal{S} = \mathbf{E} \times \mathbf{B}/\mu_0 = \epsilon_0 c \mathbf{E} \times (c\mathbf{B})$.
- Parts 2-4 assume $E_{10} = E_{20} \equiv E_0$ and use $X = kx + \phi/2$, $T = \omega t + \phi/2$.

Symbols

Symbol	Meaning	Units	Assumptions
E_{10}	amplitude of wave 1 (travels toward $+\hat{x}$)	V/m	$E_{10} > 0$
E_{20}	amplitude of wave 2 (travels toward $-\hat{x}$)	V/m	$E_{20} > 0$
E_0	common amplitude when $E_{10} = E_{20}$ (parts 2-4)	V/m	$E_0 > 0$
k	wavenumber, $k = \omega/c = 2\pi/\lambda$	1/m	$k > 0$
ω	angular frequency	1/s	$\omega > 0$
ϕ	phase of wave 2 relative to wave 1 at $x = t = 0$	rad	any real value
a	$a = kx - \omega t$ (phase of wave 1)	rad	
b	$b = kx + \omega t + \phi$ (phase of wave 2)	rad	
X	$X = kx + \phi/2$ (standing-wave spatial phase, parts 2-4)	rad	
T	$T = \omega t + \phi/2$ (standing-wave temporal phase, parts 2-4)	rad	
cB_z	c times the $\hat{z}$ component of $\mathbf{B}$ (same units as E)	V/m	

Symbol	Meaning	Units	Assumptions
u	energy density $u = u_E + u_B$, $u_E = \epsilon_0 E^2/2$, $u_B = B^2/2\mu_0$	J/m ³	
S_x	$\hat{x}$ component of the Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$	W/m ²	

Spot checks. Spot checks give the phases kx and ωt (radians) and ϕ directly, so you never need numerical values of k or ω . Magnetic fields are quoted as cB (in V/m); energy densities as $u/(\epsilon_0 E_0^2)$; the Poynting vector as $S_x/(\epsilon_0 c E_0^2)$. Components not listed are zero.

Part (1)

- Find B_1 and B_2 and the total B . Show that $|\mathbf{B}| \neq |\mathbf{E}|/c$ in general: find a point and time where $\mathbf{E} = 0$ but $\mathbf{B} \neq 0$, and vice versa.

Spot checks (relative tolerance 0.001):

Check	Inputs	Your result should be
1.S1	E10 [V/m] = 0.853, E20 [V/m] = 1.848, phi [rad] = 0.6166, kx [rad] = 4.617, omega t [rad] = 5.371	E = (0, -0.0824533, 0); cB = (0, 0, 1.32605)
1.S2	E10 [V/m] = 2.899, E20 [V/m] = 2.179, phi [rad] = 3.806, kx [rad] = 0.4034, omega t [rad] = 0.8686	E = (0, 3.36996, 0); cB = (0, 0, 1.81189)
1.S3	E10 [V/m] = 1.556, E20 [V/m] = 2.455, phi [rad] = 5.589, kx [rad] = 6.062, omega t [rad] = 0.9066	E = (0, 3.1219, 0); cB = (0, 0, -1.78791)

Other checks:

- 1.P1** (Special case). Example of $\mathbf{E} = 0$, $\mathbf{B} \neq 0$: with $E_{20} = E_{10}$ and $\phi = 0$, at $kx = \pi/2$ the total $E_y = 0$ for every t ...
- 1.P2** (Special case). ... while there $cB_z = 2E_{10} \sin \omega t$, which is nonzero except at isolated instants. (With $\hat{x} \times \mathbf{E}_2/c$ for wave 2 you would get $\mathbf{B} = 0$ there too.)
- 1.I1** (Identity). Faraday's law, $\partial_t B_z = -\partial_x E_y$, holds for your *total* fields (use $\omega = ck$). It is linear, so it must hold wave by wave as well.
- 1.I2** (Identity). The Ampère-Maxwell law in vacuum, $\partial_x B_z = -c^{-2} \partial_t E_y$, holds as well; together the two fix the sign of each B_j relative to its direction of travel.
- 1.I3** (Identity). $E_y + cB_z = 2E_{10} \cos a$ (wave 2 cancels): so wherever $\mathbf{E} = 0$, $cB_z = 2E_{10} \cos a$, and wherever $\mathbf{B} = 0$, $E_y = 2E_{10} \cos a$.
- 1.I4** (Identity). $E_y - cB_z = 2E_{20} \cos b$ (wave 1 cancels).
- 1.I5** (Identity). $c^2 B^2 - E^2 = -4E_{10}E_{20} \cos a \cos b$, which is not identically zero: $|\mathbf{B}| \neq |\mathbf{E}|/c$ in general (for a single travelling wave the left side vanishes).

- **1.T1** (Structure). $\mathbf{B}_1 = (E_{10}/c) \cos a \hat{z} = \hat{x} \times \mathbf{E}_1/c$ and $\mathbf{B}_2 = (-\hat{x}) \times \mathbf{E}_2/c$: each wave's $\mathbf{E} \times \mathbf{B}$ points along its own direction of travel. Both $\mathbf{B}$'s are along $\hat{z}$.
- **1.T2** (Structure). Points with $\mathbf{E} = 0$, $\mathbf{B} \neq 0$ (and vice versa) exist for *any* amplitudes E_{10}, E_{20} : at fixed t , $E_y(x)$ is a sinusoid in x and has zeros; at a zero of E_y , $cB_z = 2E_{10} \cos a \neq 0$ unless $\cos a = 0$ there as well. (An older wording, ' E maximal with $B = 0$ ', holds only for equal amplitudes.)

Wrong turns (the check in brackets catches it):

- **1.W1**: using $\hat{x} \times \mathbf{E}_2/c$ for the backward wave (wave 2 travels toward $-\hat{x}$); this makes $c\mathbf{B}$ equal to $\mathbf{E}$ rotated, puts the B -nodes on the E -nodes later, and gives $S \propto E^2 \geq 0$ [1.S1, 1.S2, 1.S3, 1.P2, 1.I1, 1.I2, 1.I3, 1.I4, 1.I5, 1.T1]
- **1.W2**: getting both $\mathbf{B}_1$ and $\mathbf{B}_2$ with the wrong overall sign (check $\mathbf{E}_1 \times \mathbf{B}_1$ must point toward $+\hat{x}$) [1.S1, 1.S2, 1.S3, 1.I1, 1.I2, 1.I3, 1.I4, 1.T1]
- **1.W3**: putting $\mathbf{B}$ along $\hat{y}$ (parallel to $\mathbf{E}$) instead of $\hat{z}$ [1.S1, 1.S2, 1.S3, 1.T1]
- **1.W4**: copying the phase $a = kx - \omega t$ into wave 2 instead of using $b = kx + \omega t + \phi$ [1.S1, 1.S2, 1.S3, 1.I1, 1.I2, 1.I3, 1.I4]

Part (2)

2. What condition on E_{10} and E_{20} makes the total electric field a standing wave, i.e. of the form $f(x)g(t)$? Assume it from now on. Write $\mathbf{E}$ and $\mathbf{B}$ in standing-wave form and find the nodes of each for arbitrary ϕ . By how much are the E -nodes and B -nodes displaced from one another in space?

Spot checks (relative tolerance 0.001):

Check	Inputs	Your result should be
2.S1	E_0 [V/m] = 1.02, ϕ [rad] = 2.97, kx [rad] = 1.919, ωt [rad] = 1.167	$\mathbf{E} = (0, 1.73872, 0)$; $c\mathbf{B} = (0, 0, -0.24886)$
2.S2	E_0 [V/m] = 2.285, ϕ [rad] = 0.4153, kx [rad] = 5.957, ωt [rad] = 4.872	$\mathbf{E} = (0, 1.62939, 0)$; $c\mathbf{B} = (0, 0, 0.504399)$
2.S3	E_0 [V/m] = 1.556, ϕ [rad] = 4.872, kx [rad] = 2.531, ωt [rad] = 6.021	$\mathbf{E} = (0, -0.444527, 0)$; $c\mathbf{B} = (0, 0, -2.4805)$

Other checks:

- **2.P1** (Special case). At $X = \pi/2$ (i.e. $kx = \pi/2 - \phi/2$) $E_y = 0$ for all t : an E -node, for every ϕ .
- **2.P2** (Special case). At that same E -node, $cB_z = 2E_0 \sin T$: a B -antinode.

- **2.P3** (Special case). At $X = 0$ (i.e. $kx = -\phi/2$) $cB_z = 0$ for all t : a B -node, one quarter wavelength from the E -node above.
- **2.Y1** (Symmetry). Reflecting about an E -antinode ($X \rightarrow -X$, i.e. $x \rightarrow -x - \phi/k$) leaves E_y unchanged ...
- **2.Y2** (Symmetry). ... and reverses the sign of cB_z : B is odd about the E -antinodes.
- **2.C1** (Dependence). E and B depend on x, t and ϕ only through $X = kx + \phi/2$ and $T = \omega t + \phi/2$: changing ϕ just shifts the whole pattern by $-\phi/2k$ in x and $-\phi/2\omega$ in t .
- **2.I1** (Identity). Your product form must equal the sum form: $2E_0 \cos X \cos T = E_0 \cos a + E_0 \cos b$ for all x, t, ϕ (sum-to-product identity with $X = (a + b)/2, T = (b - a)/2$).
- **2.I2** (Identity). Likewise $2E_0 \sin X \sin T = E_0 \cos a - E_0 \cos b$.
- **2.T1** (Structure). Standing wave requires $E_{10} = E_{20}$. For $E_{10} \neq E_{20}$ the sum is a standing wave of amplitude $2E_{20}$ plus a travelling wave of amplitude $E_{10} - E_{20}$ (or the reverse), not of the form $f(x)g(t)$.
- **2.T2** (Structure). E -nodes at $X = \pi/2 + m\pi$, B -nodes at $X = m\pi$: they alternate every $\lambda/4$, so the E - and B -nodes are displaced by $\lambda/4$ from each other.

Wrong turns (the check in brackets catches it):

- 2.W1: carrying the $+\hat{x} \times \underline{E}_2/c$ error of part 1 forward: $c\mathbf{B}$ comes out with the same $\cos X \cos T$ form as $\underline{E}$, so the B -nodes sit on the E -nodes [2.S1, 2.S2, 2.S3, 2.P2, 2.P3, 2.Y2, 2.I2, 2.T2]
- 2.W2: dropping or misplacing the $\phi/2$ shifts (putting all of ϕ into the time factor, or into the space factor, or dropping it) [2.S1, 2.S2, 2.S3, 2.P1, 2.P3, 2.Y1, 2.C1, 2.I1, 2.I2]
- 2.W3: overall sign error in $\mathbf{B}$ (equivalent to $\underline{B}_1$ along $-\hat{z}$) [2.S1, 2.S2, 2.S3, 2.I2]
- 2.W4: believing $\phi = 0$ is needed for a standing wave; ϕ only translates the pattern in x and t [2.C1, 2.T1]

Part (3)

3. Compute $u_E(x, t)$ and $u_B(x, t)$. Show that they are offset by a quarter wavelength in space and a quarter period in time, so that energy sloshes between electric and magnetic form. Find the total $u(x, t)$ and its time average, and compare the latter with the co-propagating result.

Spot checks (relative tolerance 0.001):

Check	Inputs	Your result should be
3.S1	ϕ [rad] = 0.9182, kx [rad] = 2.22, ωt [rad] = 1.27	$u_E/(\epsilon_0 E_0^2) = 0.0398081$; $u_B/(\epsilon_0 E_0^2) = 0.388259$; $u/(\epsilon_0 E_0^2) = 0.428067$

Check	Inputs	Your result should be
3.S2	ϕ [rad] = 5.963, kx [rad] = 1.376, ωt [rad] = 5.853	$u_E/(\epsilon_0 E_0^2) = 0.16668$; $u_B/(\epsilon_0 E_0^2) = 0.544776$; $u/(\epsilon_0 E_0^2) = 0.711456$
3.S3	ϕ [rad] = 5.952, kx [rad] = 5.596, ωt [rad] = 1.892	$u_E/(\epsilon_0 E_0^2) = 0.0207928$; $u_B/(\epsilon_0 E_0^2) = 1.1071$; $u/(\epsilon_0 E_0^2) = 1.1279$

Other checks:

- **3.P1** (Special case). At an E -antinode ($X = 0$): $u = \epsilon_0 E_0^2 [1 + \cos 2T] = 2\epsilon_0 E_0^2 \cos^2 T$, purely electric, oscillating between 0 and $2\epsilon_0 E_0^2$.
- **3.P2** (Special case). Halfway between a node and an antinode ($X = \pi/4$): $u = \epsilon_0 E_0^2$, constant in time; energy flows through this point but does not accumulate there.
- **3.Y1** (Symmetry). u is even about every E -antinode and every B -antinode ($X \rightarrow -X$).
- **3.Y2** (Symmetry). u is even in time about the instants of maximal E ($T \rightarrow -T$).
- **3.C1** (Dependence). u_E , u_B , u depend on x, t, ϕ only through X and T .
- **3.I1** (Identity). $u_E(x, t) = u_B(x + \lambda/4, t + \tau/4)$: the magnetic pattern is the electric one shifted by a quarter wavelength and a quarter period ($\tau = 2\pi/\omega$).
- **3.I2** (Identity). $u = \epsilon_0 E_0^2 [1 + \cos(2kx + \phi) \cos(2\omega t + \phi)]$.
- **3.I3** (Identity). Time average over a period: $\langle u \rangle = \epsilon_0 E_0^2$, the same at every x and for every ϕ .
- **3.I4** (Identity). For comparison, two co-propagating waves of the same amplitude E_0 (problem 6) have $\langle u \rangle = \epsilon_0 E_0^2 (1 + \cos \phi)$, anywhere between 0 and $2\epsilon_0 E_0^2$ depending on ϕ ; the standing wave always has the ϕ -independent value $\epsilon_0 E_0^2$, the incoherent sum of the two waves' averages.
- **3.T1** (Structure). $u_E \propto \cos^2 X \cos^2 T$ and $u_B \propto \sin^2 X \sin^2 T$ with the same prefactor $2\epsilon_0 E_0^2$: at an E -antinode the energy is all electric at $T = 0$ and all gone a quarter period later, when it sits in the magnetic field a quarter wavelength away.

Wrong turns (the check in brackets catches it):

- 3.W1: taking $u = \epsilon_0 E^2/2$ as the *total* energy density (that is u_E alone; in a standing wave $u_B \neq u_E$, so you cannot just double it either) [3.S1, 3.S2, 3.S3, 3.P2, 3.I2, 3.I3, 3.T1]
- 3.W2: using the wrong-sign B_2 from part 1, which makes $u_B = u_E$ everywhere and $u = \epsilon_0 E^2$ with no sloshing [3.S1, 3.S2, 3.S3, 3.P1, 3.P2, 3.I1, 3.I2, 3.I3, 3.T1]
- 3.W3: dropping the $\phi/2$ shifts, so the ϕ -dependence of u comes out wrong (it must enter only as $\cos(2kx + \phi) \cos(2\omega t + \phi)$) [3.S1, 3.S2, 3.S3, 3.C1, 3.I2]
- 3.W4: comparing $\langle u \rangle$ with a ϕ -dependent co-propagating value without saying the amplitudes are equal there too [3.I4]

Part (4)

4. Compute $\mathcal{S}(x, t)$ and its time average. Where does $\mathcal{S}$ point at a given instant, and why is its time average zero everywhere? Reconcile this with the fact that u oscillates locally.

Spot checks (relative tolerance 0.001):

Check	Inputs	Your result should be
4.S1	ϕ [rad] = 0.9137, kx [rad] = 6.021, ωt [rad] = 1.387	$S_x/(\epsilon_0 c E_0^2) = -0.197134$
4.S2	ϕ [rad] = 5.884, kx [rad] = 0.3697, ωt [rad] = 5.642	$S_x/(\epsilon_0 c E_0^2) = -0.331645$
4.S3	ϕ [rad] = 0.9502, kx [rad] = 0.393, ωt [rad] = 2.843	$S_x/(\epsilon_0 c E_0^2) = 0.34101$

Other checks:

- **4.P1** (Special case). $S_x = 0$ at every B -node ($X = m\pi$) for all t ...
- **4.P2** (Special case). ... and at every E -node ($X = \pi/2 + m\pi$): energy never crosses a node, so each $\lambda/4$ cell between an E -node and a B -node is energetically closed.
- **4.P3** (Special case). At $X = \pi/4$ (where u is constant in time), $S_x = \epsilon_0 c E_0^2 \sin 2T$: the flow is largest where the energy density does not change.
- **4.Y1** (Symmetry). S_x is odd about every node and antinode ($X \rightarrow -X$): at a given instant it points away from (or toward) an E -antinode symmetrically on both sides.
- **4.Y2** (Symmetry). S_x is odd in time about the instants of maximal E ($T \rightarrow -T$): the flow reverses every quarter period.
- **4.C1** (Dependence). S_x depends on x, t, ϕ only through X and T .
- **4.I1** (Identity). $S_x = \epsilon_0 c E_0^2 \sin(2kx + \phi) \sin(2\omega t + \phi)$ (double-angle identities).
- **4.I2** (Identity). Energy conservation holds locally: $\partial_t u + \partial_x S_x = 0$ with your u from part 3 and your S_x , once $\omega = ck$ is used. (Try it: a sign error in B breaks it.)
- **4.I3** (Identity). $\langle S_x \rangle = 0$ at every x : no net energy transport.
- **4.T1** (Structure). At a given instant $\mathcal{S}$ alternates in direction from one $\lambda/4$ cell to the next: energy flows from the E -antinodes toward the B -antinodes while u_E decreases, and back a quarter period later. $\langle S_x \rangle$ is independent of x (since $\partial_x \langle S_x \rangle = -\langle \partial_t u \rangle = 0$) and vanishes at the nodes, hence vanishes everywhere; a time-oscillating u with zero mean flow is consistent because $\partial_t u = -\partial_x S_x$ averages to zero over a period.

Wrong turns (the check in brackets catches it):

- 4.W1: carrying the $+\hat{x} \times \underline{E}_2/c$ error forward: $\mathcal{S} \propto E^2 \geq 0$ always points toward $+\hat{x}$ and has a nonzero average [4.S1, 4.S2, 4.S3, 4.P1, 4.P2, 4.P3, 4.Y1, 4.Y2, 4.I1, 4.I2, 4.I3, 4.T1]
- 4.W2: reusing the co-propagating result $\mathcal{S} = cu \hat{x}$, which does not hold here (there is no single $\hat{k}$) [4.S1, 4.S2, 4.S3, 4.P1, 4.P2, 4.P3, 4.Y1, 4.Y2, 4.I1, 4.I2, 4.I3]
- 4.W3: dropping the $\phi/2$ shifts [4.S1, 4.S2, 4.S3, 4.P1, 4.P2, 4.C1, 4.I1]

- 4.W4: overall sign error (from a sign error in B) [4.S1, 4.S2, 4.S3, 4.I2]